

GE261004 DESIGN THINKING

Unit IV – Prototyping and Testing · Presentation 2 of 6

Sketching and Mock-ups

Why engineers should sketch · five marks that draw anything · thumbnails, sketches and mock-ups · storyboarding · paper mock-ups

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In this presentation

- Why engineers should sketch
- Five marks are enough
- Thumbnail, sketch, mock-up
- In class: ten sketches
- What a storyboard is for
- Building a storyboard
- What storyboards catch
- Paper mock-ups



1

Sketching

A thinking tool, not an art form

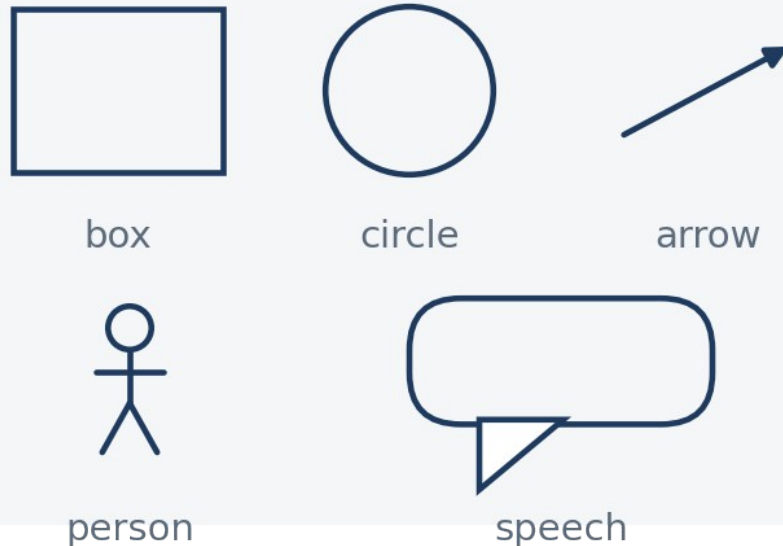
Why engineers should sketch

- A sketch is the fastest way to get an idea out of your head and onto a table
- It is ambiguous on purpose – which invites other people to change it
- Ten rough sketches beat one careful drawing, because you can compare them
- Nobody is judging the drawing; they are judging the idea it carries

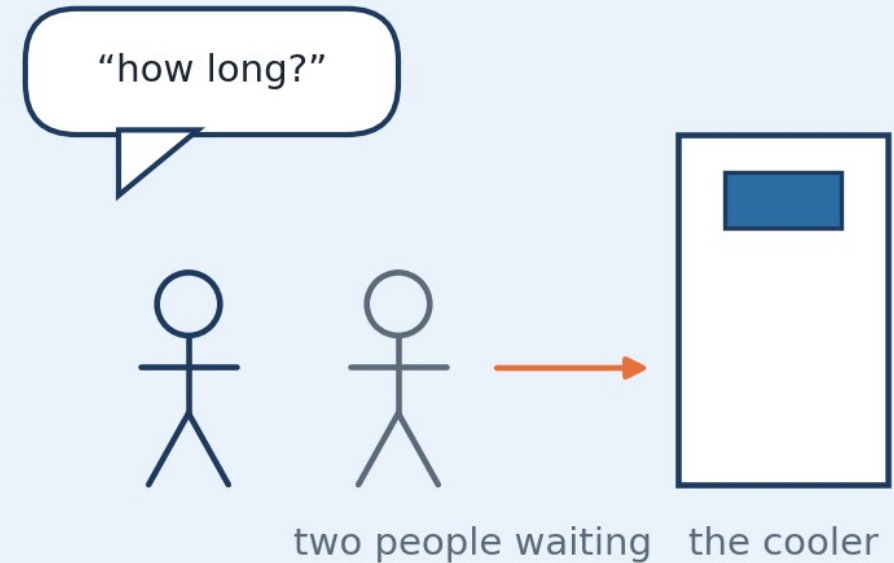
KEY POINT The skill you need is not drawing well. It is being willing to be seen drawing badly.

Five marks are enough

Five marks are enough



and they combine into a scene



A box, a circle, an arrow, a stick figure and a speech bubble.

How to sketch an idea

- Draw the user first, then the thing – it forces you to show the situation
- Use arrows for motion and speech bubbles for what people say
- Label anything ambiguous with two words; do not draw it more carefully
- Put a title on every sketch, or in a week nobody will know what it was

ACTIVITY Ten minutes, ten sketches, one per minute. Same idea each time, drawn differently. No erasing.

Thumbnail, sketch, mock-up

Thumbnail

Postage-stamp size, seconds each. For comparing many options at a glance.

Sketch

Half a page, a minute or two. For explaining one option to another person.

Mock-up

Made of paper or card, at real size. For putting in somebody's hands.

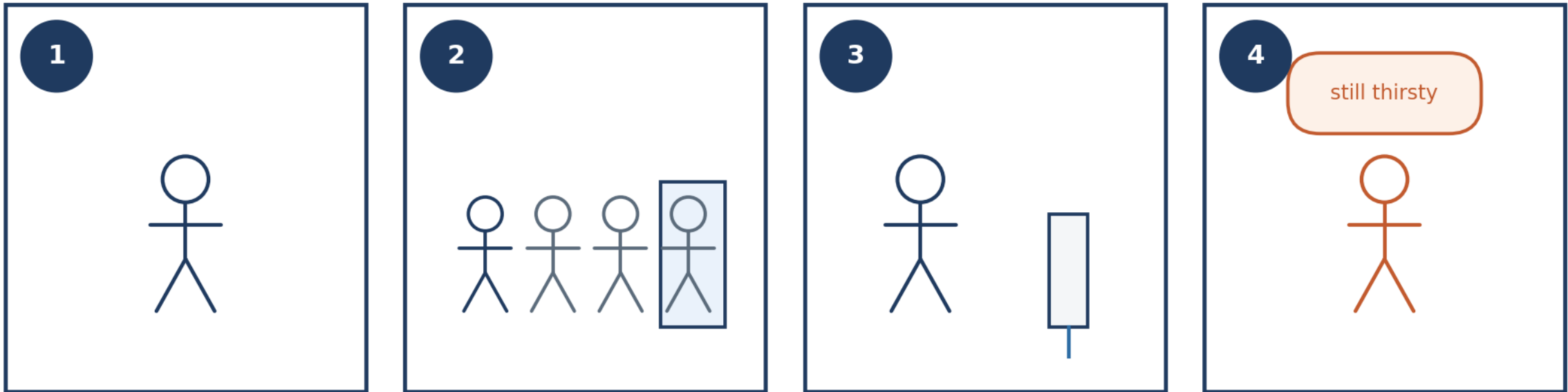
KEY POINT Start at thumbnail size. Small drawings stop you polishing something you have not chosen yet.

2

Storyboarding

Drawing the experience, not the object

A four-panel storyboard



She arrives between
classes, thirsty

The queue is eight
people long

She refills at a tap
on the way instead

Two hours later she
still has not drunk

four panels: the need, the obstacle, the workaround, the consequence

The same problem told as a sequence, with a person in every frame.

Building a storyboard

- Four to eight panels – more than that and nobody reads it
- A person in every panel; an object alone tells you nothing
- One caption per panel, in the present tense, describing what happens
- Include the moment before and the moment after, not only the solution working

KEY POINT If your storyboard has no panel where something goes wrong, you have drawn an advertisement, not a design.

What a storyboard catches

- Steps you forgot – finding it, getting to it, carrying it, cleaning it
- People you forgot: who hands it over, who maintains it, who is waiting behind
- Time: how long each step takes, and where the waiting happens
- The moment the user gives up, which is the moment worth designing for

KEY POINT A product works in a demonstration. A storyboard shows whether it works in a day.

Paper mock-ups

- Cut card to the real size and weight – scale is a thing you cannot imagine
- Draw the screens or panels by hand; printing them makes people too polite
- Hand it to somebody and ask them to do a task with it, out loud
- Change it between users with scissors and a marker, during the session

KEY POINT A mock-up you can cut up between two users has done more work than a model you had to book a lab to make.

In class: storyboard your concept

- Take the concept your team chose in Unit III
- Draw four panels: the need, the obstacle, your solution in use, the result
- Put a person in every panel and one caption under each
- Swap with another team and have them tell you the story back

ACTIVITY Twenty minutes. If the other team tells a different story, the storyboard is the thing that needs fixing.

THINK ABOUT IT

Which panel of your storyboard did you find hardest to draw?

- Was it hard because you cannot draw, or because you do not know what happens?
- Which step did you skip because it was boring?
- Who appears in the story that you had not thought about?
- Where does your user give up?

Summary

- Sketching is a thinking tool; ambiguity invites other people in
- Box, circle, arrow, stick figure, speech bubble – that is the whole kit
- Thumbnails compare, sketches explain, mock-ups get handled
- A storyboard draws the experience over time, with a person in every panel
- It catches forgotten steps, forgotten people and the moment of giving up
- Paper mock-ups can be cut up and changed between two users

Next: Low-fidelity prototyping

- What low fidelity buys you
- Paper, cardboard, clay and role play
- Wizard of Oz: testing behaviour before it exists
- Building something usable inside one hour